

Chapter 1

Introduction to Digital Image Processing



What is Digital Image Processing?

□ Image:

An image may be defined as a two dimensional function, $f(x,y)$, where x and y are spatial (plane) coordinates, and the amplitude of f at any pair of coordinates (x,y) is called the **intensity**, or **gray level** of the image at that point. When (x,y) and f are all finite and discrete quantities, we call the image a **digital image**.

□ Digital Image Processing (DIP):

It refers to the processing of digital image by means of a digital computer.

Image Processing, Image Analysis and Computer Vision

- ❑ Broadly image processing consists of three levels of processes: Low, Middle, and High.
- ❑ The **low level** processes involves those operations whose inputs are images and whose outputs are also images. For example, image enhancement operations such as image negative and smoothing of an image fall in this category. These operations are the mainstay (backbone) of the **DIP**.
- ❑ The **mid level** processes are characterized by the fact that its inputs generally are images, but its outputs are attributes extracted from these images, e.g. edges, contours and the identity of individual objects.

Mid level processing on images involves tasks such as:

- ❖ **Segmentation** (partitioning an image into regions or objects),
- ❖ **Description** of those objects (in terms of features extracted) to reduce them to a form suitable for computer processing and
- ❖ **Classification** (recognition) of individual objects.

Image Processing, Image Analysis and Computer Vision

- ❑ The **higher level** processing involves “making sense” of an ensembles (group) of recognized objects and performing the cognitive functions normally associated with vision.
- ❑ There is very little difference between **DIP and Image Analysis and Computer Vision**.
- ❑ **DIP** covers both low and mid level processing till recognition of individual regions or objects in the image.
- ❑ **Image Analysis** deals with the **Knowledge extraction** part out of an image (or visual input) i.e., whether the recognized objects in the image makes any sense or not (i.e. give any useful knowledge or not).
- ❑ **Computer Vision** then uses this knowledge to take decisions and make any necessary actions.
- ❑ The ultimate goal of **Computer Vision** is to use computers to emulate human vision, including learning and being able to make inferences and take actions based on visual inputs.

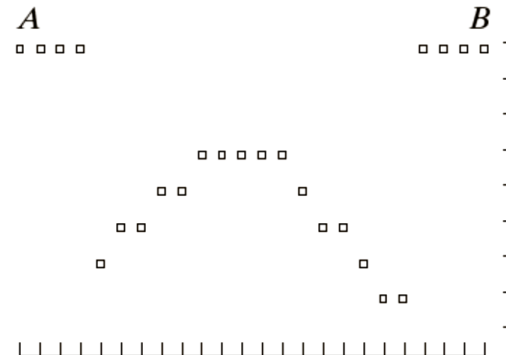
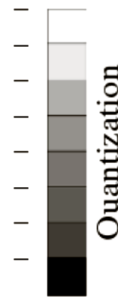
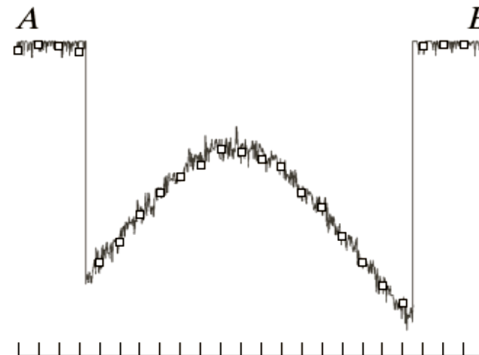
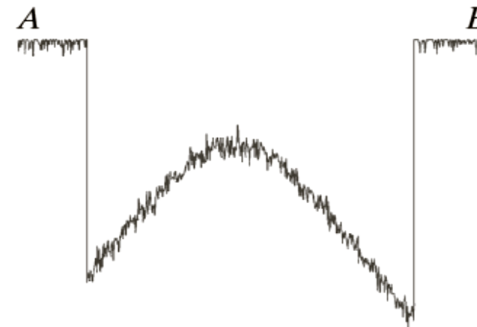
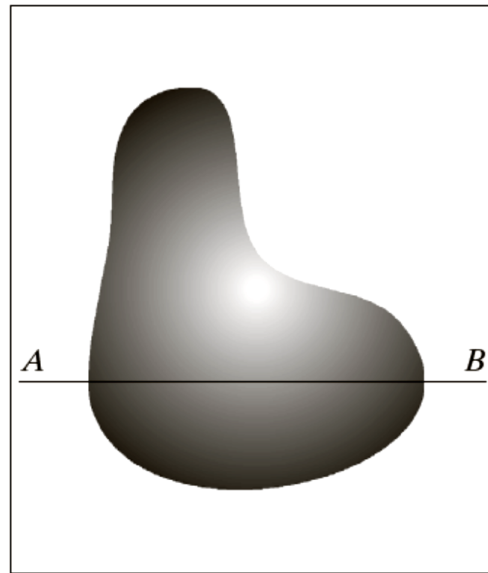
Difference between Image Processing and Computer Graphics

- ❑ **Computer Graphics:** Computer Graphics is a branch of computer science that deals with the creation, modification and manipulation of images with the help of a digital computer.
- ❑ The key element that distinguishes DIP from Computer Graphics is that the ***Image Processing generally begins with images, while the basic operation in Computer Graphics is to create images.*** While each of these two fields has its own focus and strength, they also overlap and complement each other.

Image Sampling And Quantization

- ❑ A digital image is always only an **approximation** of a real world scene.
- ❑ Digitizing the coordinate values is called **sampling**.
- ❑ Digitizing the amplitude values is called **quantization**.

Image Sampling And Quantization



Fundamental Steps in Digital Image Processing

All steps include the following

1. Image Acquisition
2. Image Filtering and Enhancement
3. Image Restoration
4. Color Image Processing
5. Wavelets and Multiresolution Processing
6. Image Compression
7. Morphological Processing
8. Image Segmentation
9. Image Representation and Description
10. Object Recognition

Fundamental Steps in Digital Image Processing

- 1. Image Acquisition:** This step pertains to the acquisition of an image agreeable to process it digitally. Whatever be the process of acquiring an image, it will finally be in a digital form which can be stored in a system in the form of a file.

Note that acquisition of an image could be as simple as being an image that is already in digital form.

- 2. Image Filtering and Enhancement:** It is the simplest and the most appealing area of digital image processing. It is process of manipulating the image so that the result is more suitable than the original one for a specific application.

The ideas behind enhancement techniques are to bring out details that are obscured, or simply to highlight certain features of interest of the image. For example: Increasing the contrast of an image.

The measure is “goodness” or “badness” depending upon the psychophysical aspect of human visual system.

Fundamental Steps in Digital Image Processing

- 3. Image Restoration:** This also deals with improving the appearance of an image. However, unlike **enhancement, which is subjective** (depends upon perception of an individual), **image restoration is objective**. Image restoration techniques tend to be based on mathematical or probabilistic models of image degradation. This approach usually involves formulating a criterion of goodness (e.g.. PSNR) that will yield an optimal estimate of the desired result.
- 4. Color Image Processing:** Color Image Processing is used to extract features of interest in color images. It is gaining importance because of increasing use of color images over the Internet.
- 5. Wavelets and Multi-resolution Analysis:** Wavelets are used to represent images in various degrees of resolution. It is also used in image compression and for pyramidal representation in which images are subdivided successively into smaller regions.

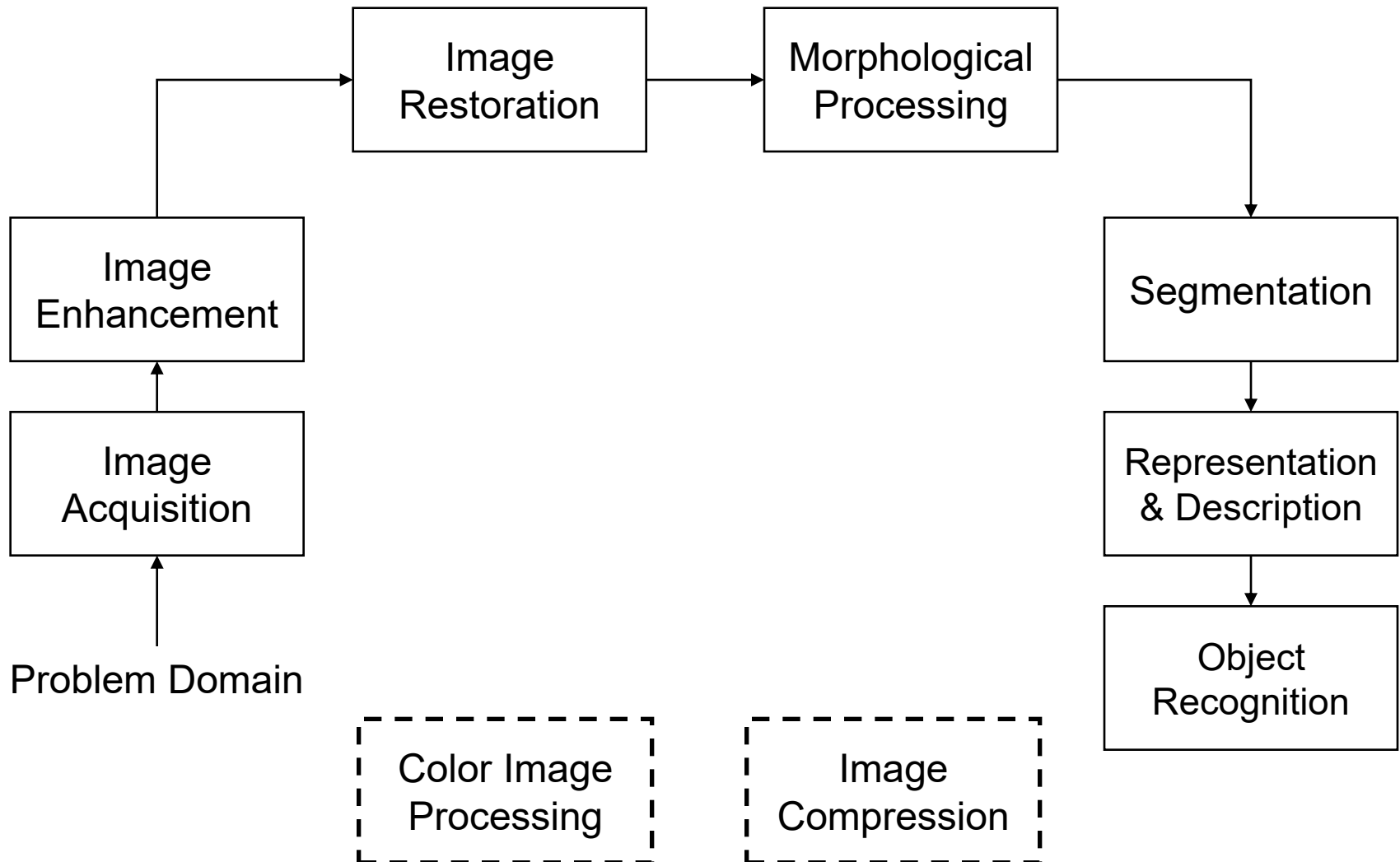
Fundamental Steps in Digital Image Processing

- 6. Image Compression:** Deals with reducing the storage requirements of an image. The use of images has increased manifold in information technology for the last few years, especially after the intensive use of the Internet. The size of image affects the transmission quality of information over Internet.
- 7. Morphological Processing:** Deals with the tools for extracting image components, which are useful in the representation and description of region shape, such as boundaries, skeletons, and the convex hull. This also includes the techniques for pre and post processing, such as morphological filtering, thinning and pruning.
- 8. Image Segmentation:** Deals with procedures for partitioning an image into its constituent parts. For example, a scanned page may be split into photographs, texts, tables and columns. In general autonomous segmentation is one of the most difficult task in digital image processing.

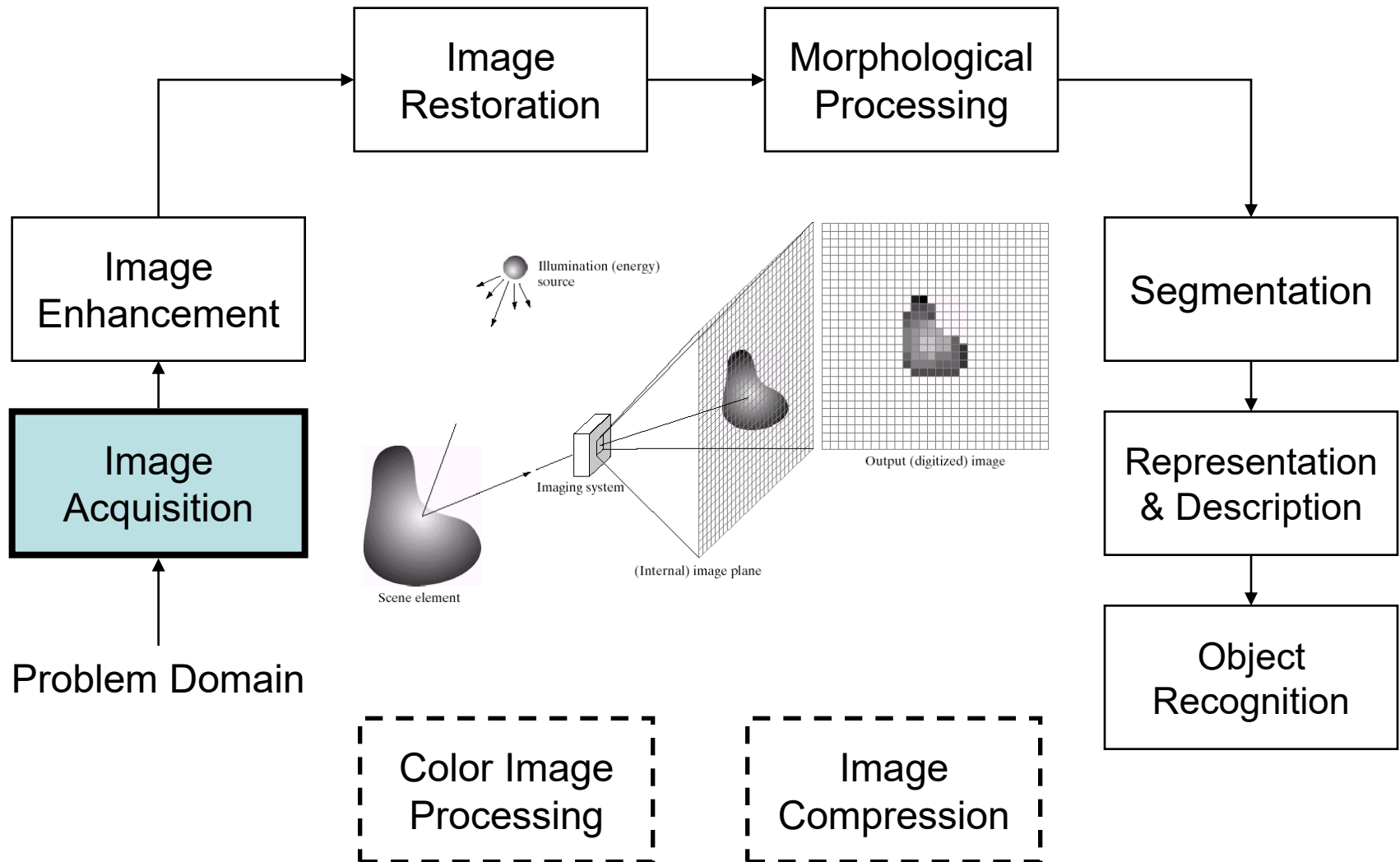
Fundamental Steps in Digital Image Processing

- 9. Image Representation and Description:** It deals with extracting attributes that result in some quantitative information of interest or are basic for differentiating one class of objects from others. This stage usually processes the data of segmentation stage. Description deals with extracting attributes that are helpful in identifying an object.
- 10. Image Recognition:** Image Recognition is a process which assigns a label (based on its descriptors) to the digital image after matching it with a known object. For example, the decision whether an image or part thereof is the human face or not.

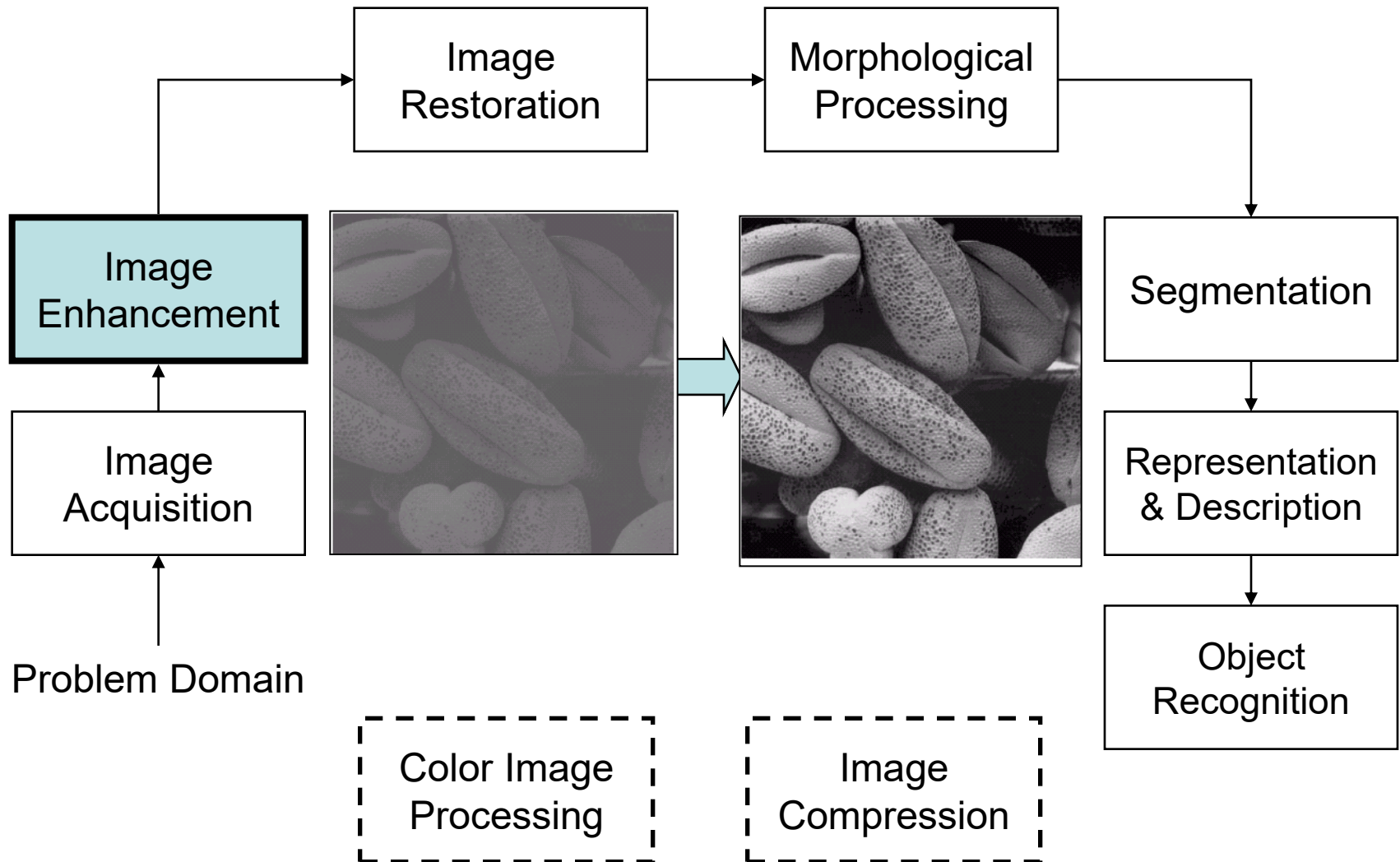
Key Stages in Digital Image Processing



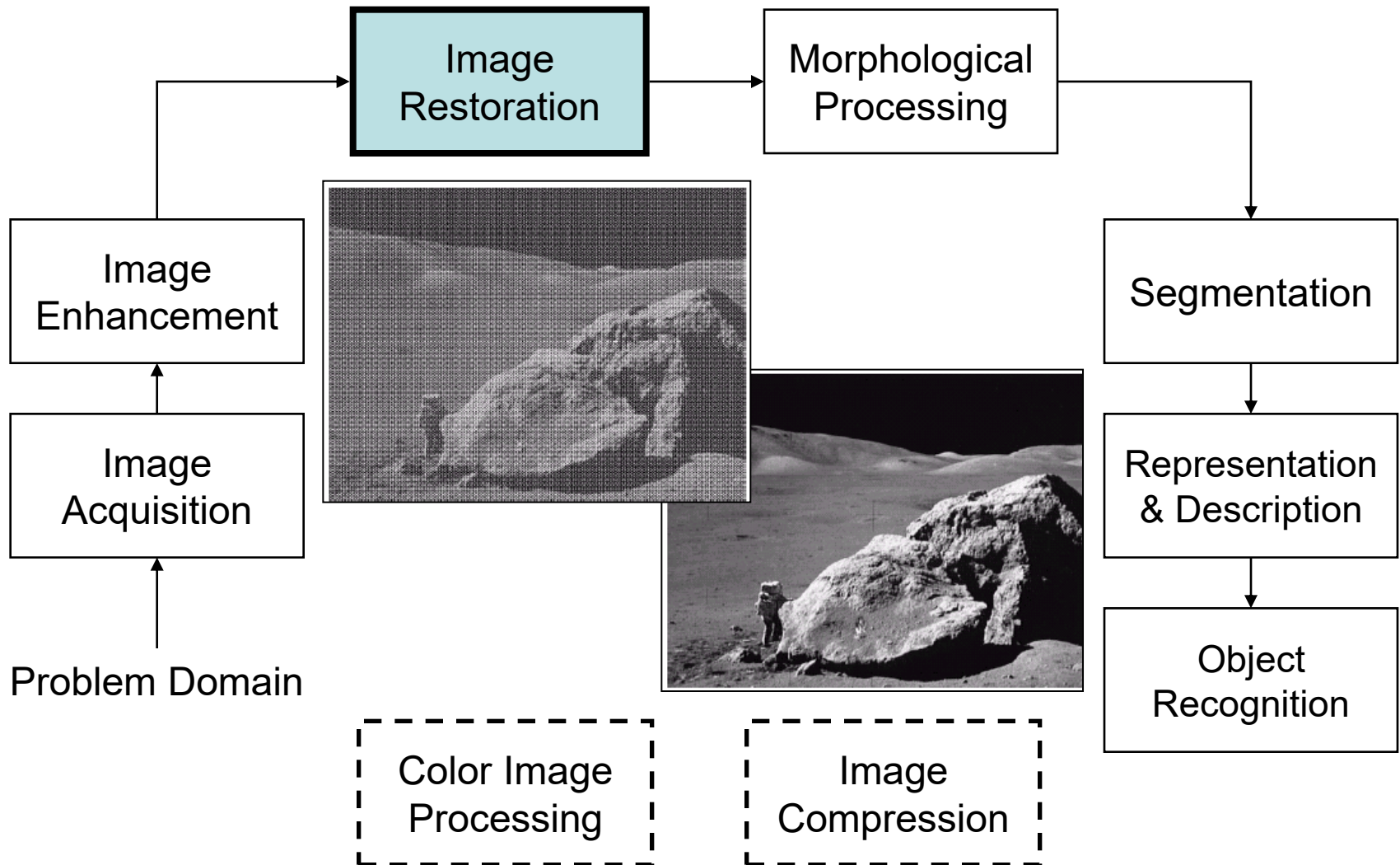
Key Stages in Digital Image Processing: Image Aquisition



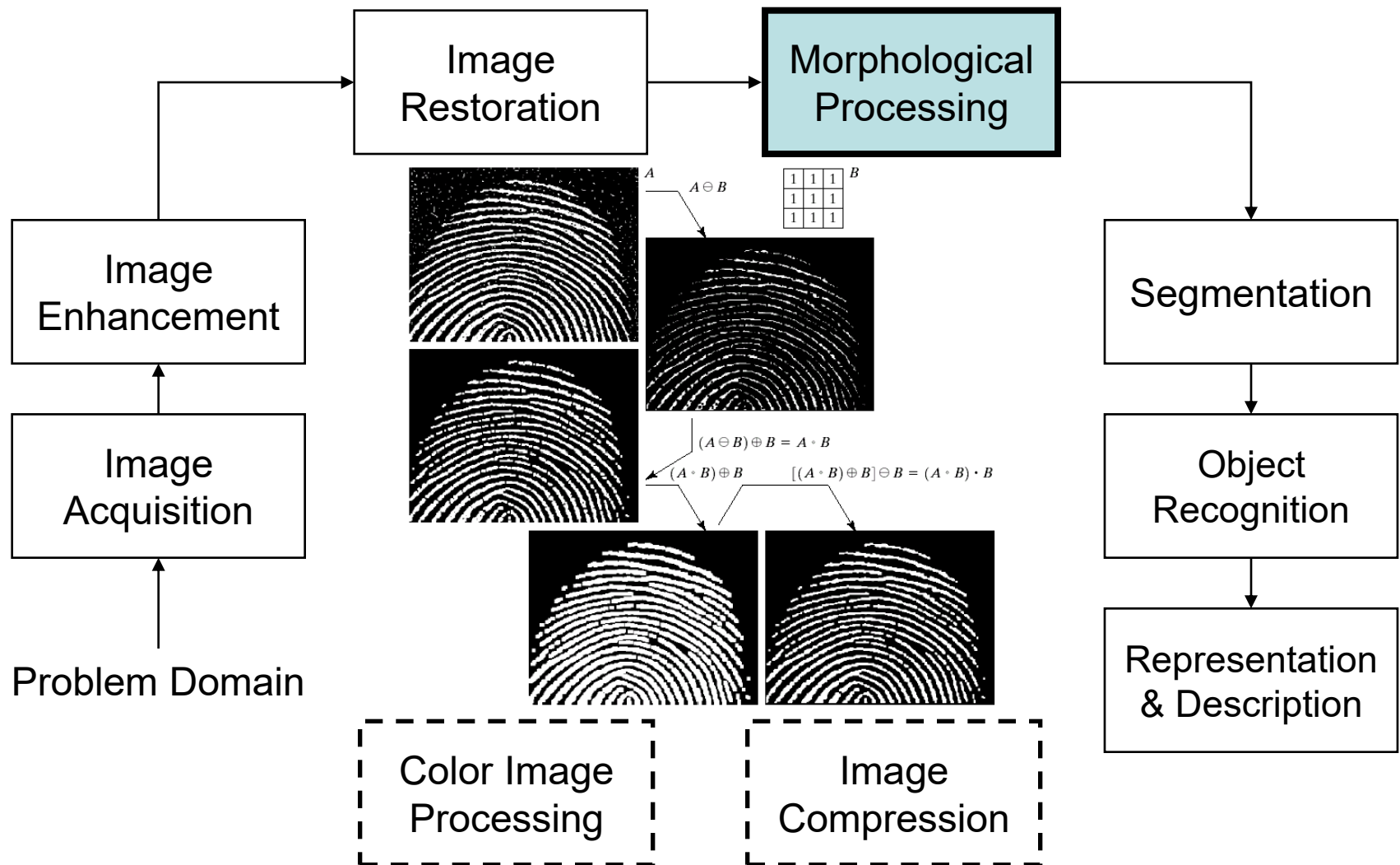
Key Stages in Digital Image Processing: Image Enhancement



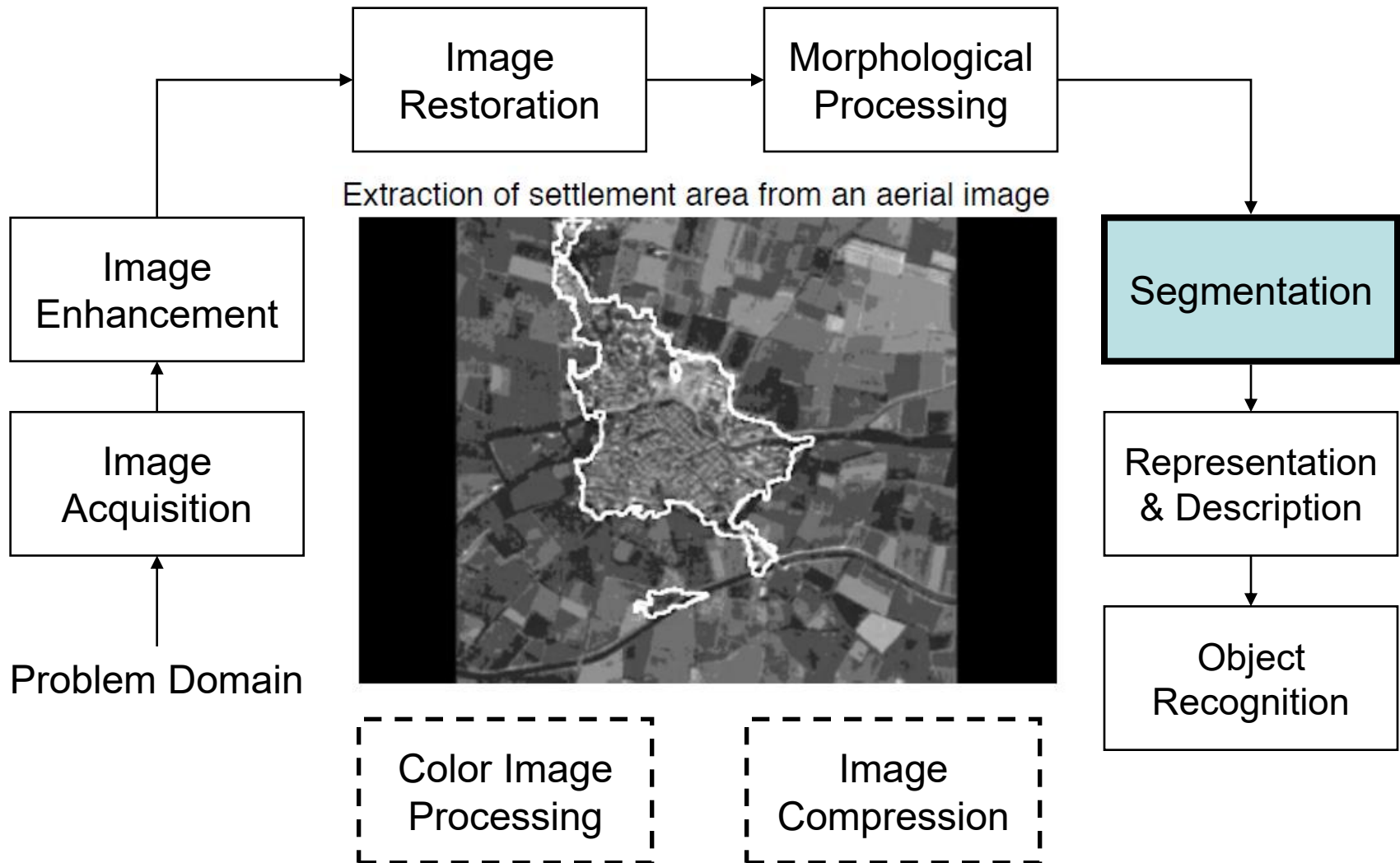
Key Stages in Digital Image Processing: Image Restoration



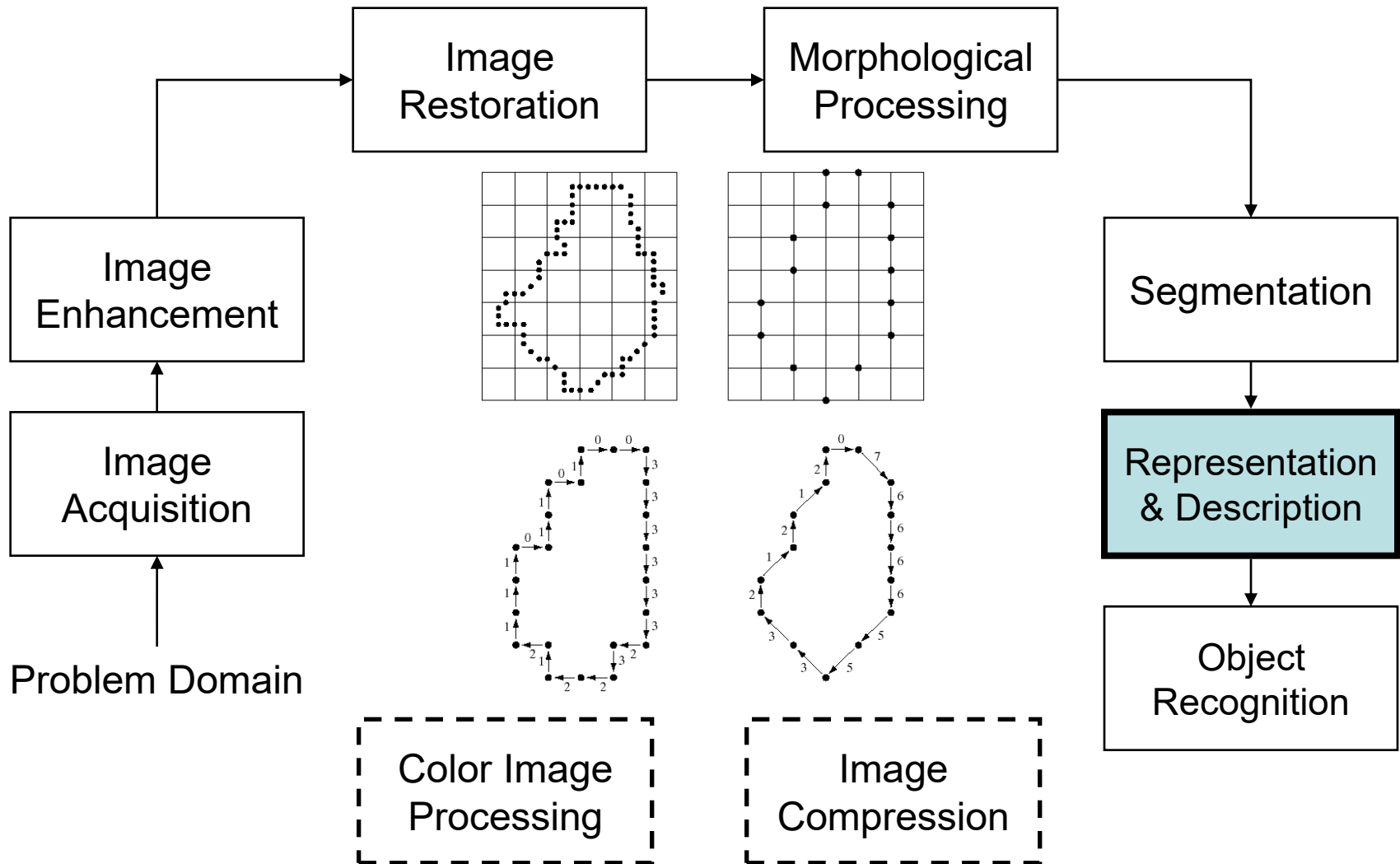
Key Stages in Digital Image Processing: Morphological Processing



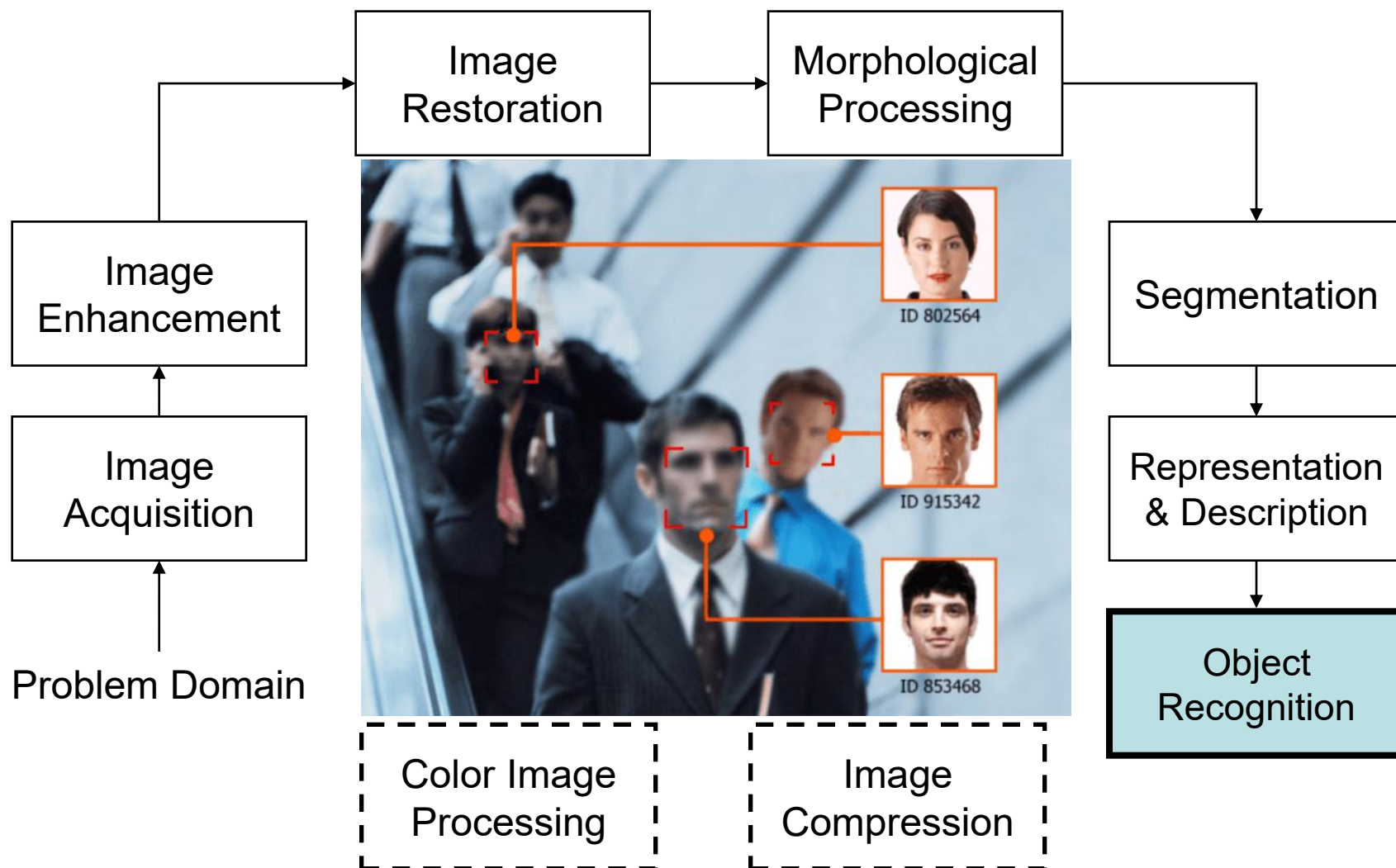
Key Stages in Digital Image Processing: Segmentation



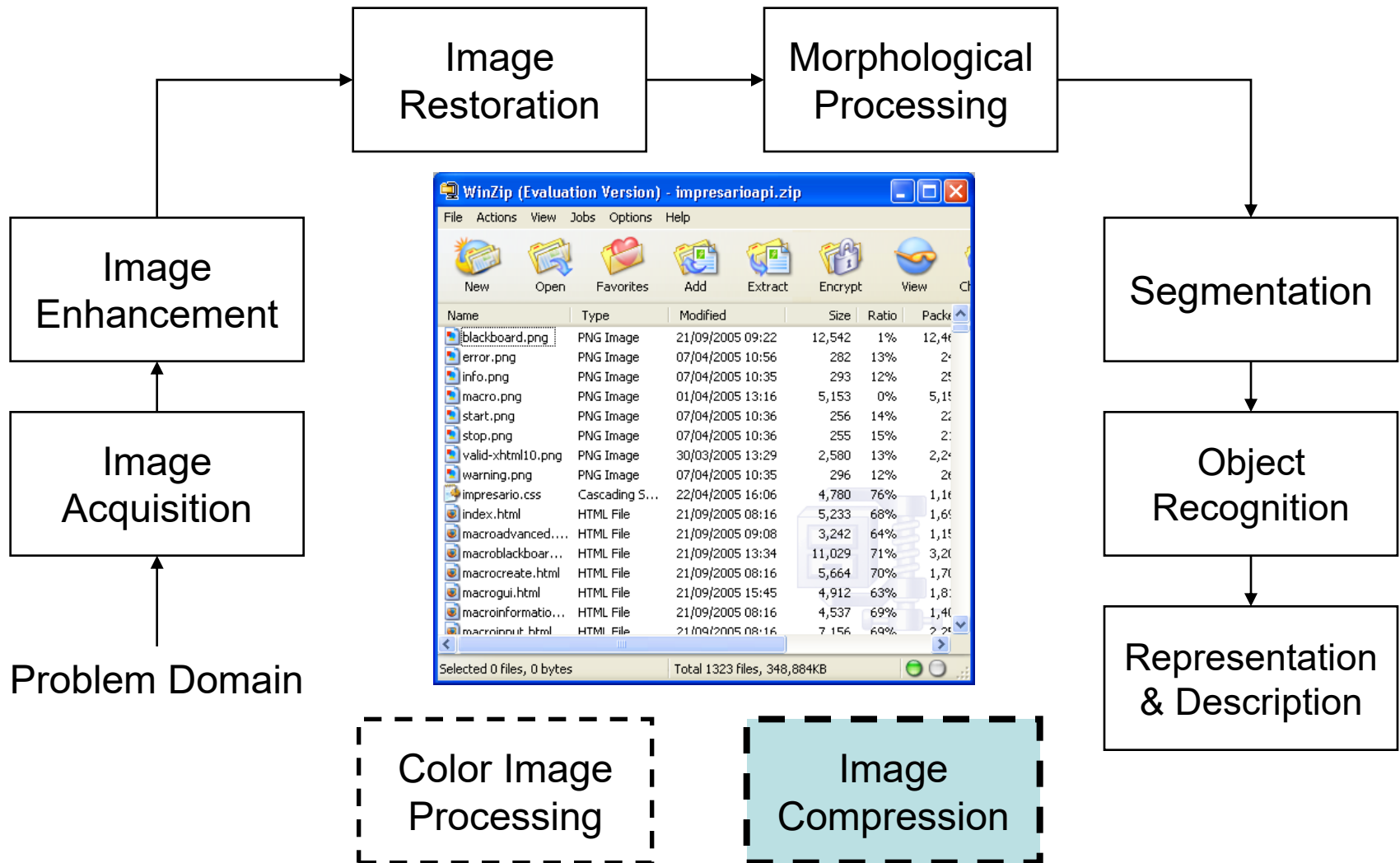
Key Stages in Digital Image Processing: Object Recognition



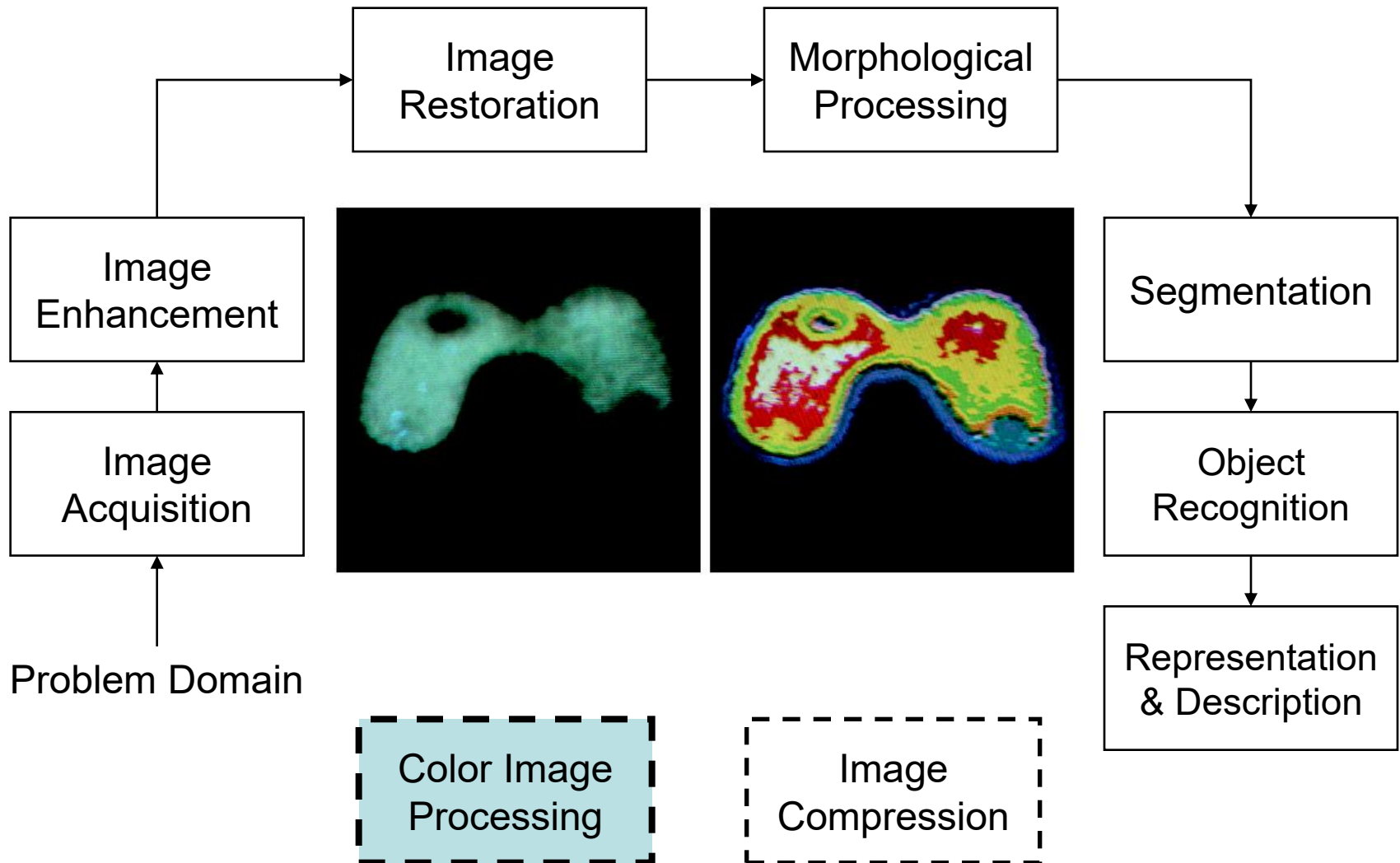
Key Stages in Digital Image Processing: Representation & Description



Key Stages in Digital Image Processing: Image Compression



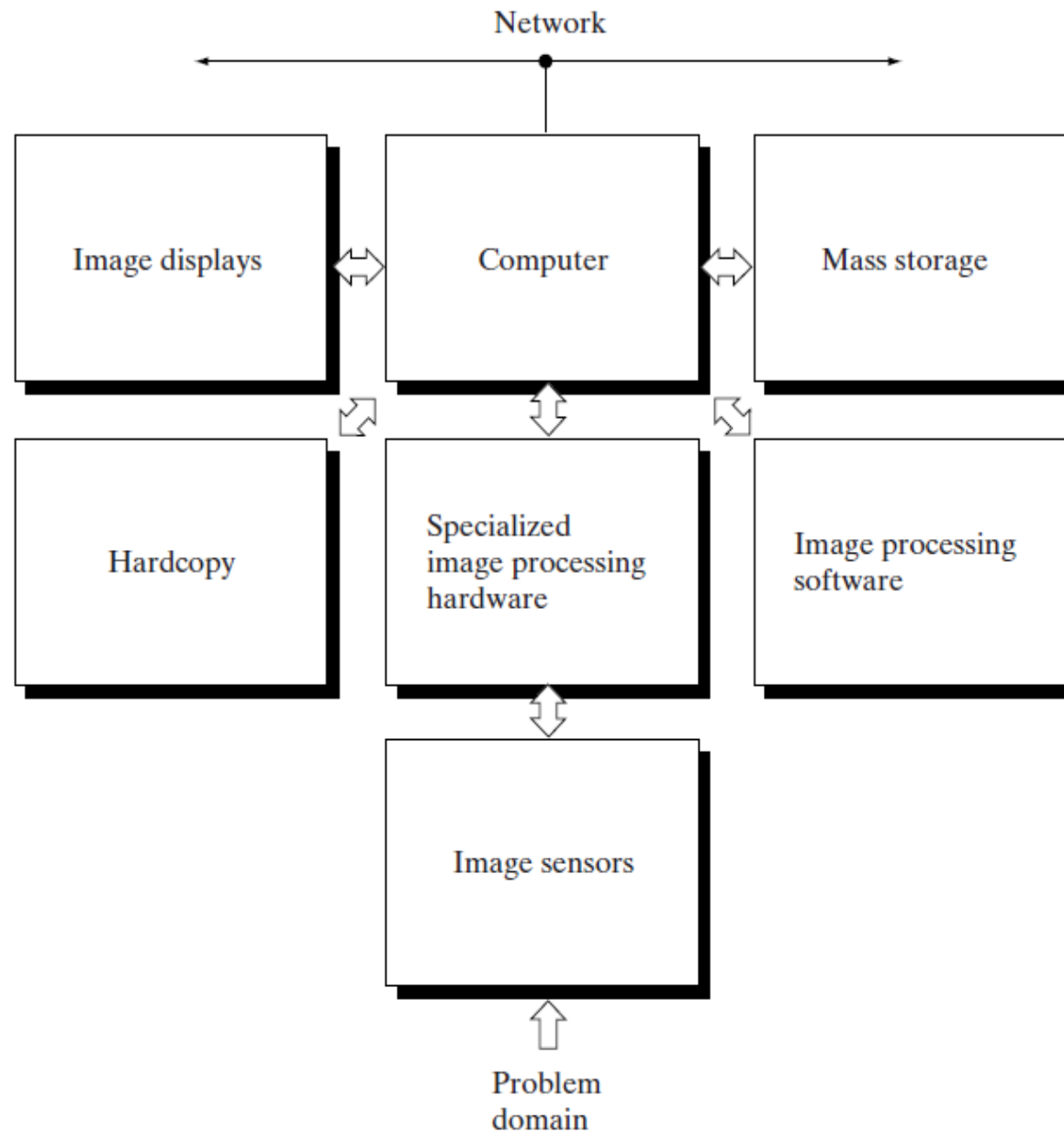
Key Stages in Digital Image Processing: Colour Image Processing



Components of an Image Processing System

An image processing system consists of a set of hardware devices and software components. The major components are:

1. Image Sensors and Digitizers
2. Specialized Image Processing Hardware or Computer
3. Image Processing Software
4. Mass storage
5. Image Display and Hardcopy Devices
6. Image Transmission Systems

**FIGURE**

Components of a general-purpose image processing system.

Components of an Image Processing System

- 1. Image Sensors and Digitizer:** Image Sensor is a physical device that is sensitive to the light radiated by the object we wish to image. A digitizer is a device that converts the analog form of the output of an image sensor into digital form. For example, in a digital video camera, the sensors produce an electrical output proportional to light intensity and the digitizer converts these outputs to digital data.
- 2. Specialized Image Processing Hardware or Computer:** Although a general purpose computer can perform the operations on an image after the related software is run on it, specialized hardware, on the other hand, can perform some of the primitive operations very fast. For example, a specialized hardware can be used to average image pixels immediately after digitizing an image for the purpose of noise reduction. The Computer performs the general purpose operations.

Components of an Image Processing System

- 3. Image Processing Software:** Software is required for image processing operations. For example, image segmentation software can segment the image into various components, an OCR system can convert a document image into editable text and it can separate out the various components of the document such as the text, tables, figures, images, etc.
- 4. Mass Storage:** Images require very large amount of storage. For example, an image of the size 1024×1024 and capable of displaying 256 gray levels requires 1MB of space. Storage may be divided into three categories:
 - ❖ *Short term storage:* Storage for use during processing (computer memory-RAM and video memory for image display).
 - ❖ *On-line Storage:* for relatively fast recall (normally a hard disk or CDs).
 - ❖ *Archival Storage:* for back up purpose (offline storage of many images) such as magnetic tapes and optical disks.

Components of an Image Processing System

- 5. Image Display and Hardcopy Devices:** The display is normally made on a video screen-monitor or computer screen. Hardcopy devices include laser and inkjet printers and plotters.
- 6. Image Transmission System:** Image transmission system has become an integral part of image processing. Images are frequently exchanged over the Internet for processing them. The key consideration in image transmission is the bandwidth.

Types of Images

❑ There are basically three types of images:

1. Binary Image
2. Gray Scale Image
3. Color Image

1. **Binary Image:** A binary image has two levels: 0 (black) and 1 (white). Only one bit is required to represent a pixel. Thus if the size of a binary image is $M \times N$, then $M \times N$ bits are needed to represent the image.
2. **Gray Scale Image:** A gray scale image has 256 gray levels, 0 to 255. A pixel is represented by 8 bits or a byte. Thus a gray scale image of the size $M \times N$ will need $M \times N$ bytes.
3. **Color Image:** A color image has more than one component and normally two representations, RGB and CMY are most common. The color representation RGB is used for video display devices and CMY is used for hard copy devices.

Each component requires one byte for its representation. Thus, a color pixel requires 3 bytes of storage. The total numbers of colors is $256 \times 256 \times 256 = 2^{24} = 16.7$ Million. An $M \times N$ size of color image will require $3 \times M \times N$ bytes of storage.

Spatial Resolution

- ❑ **Spatial Resolution:** It represents the number of pixels in an image. It is given by

$$\text{No. of pixels along width} \times \text{No. of pixels along height}$$

- ❑ Two other units for spatial resolution depending upon the number of pixels per unit length: *dots per inch (dpi)* and *pixels per inch (ppi)*.
- ❑ Formula for computing the PPI of a mobile device or any screen

$$PPI = \frac{\sqrt{w^2 + h^2}}{d}$$

where

d is the size of the screen (diagonally)

$w \times h$ is the maximum resolution supported by the screen.

- ❑ **Bit depth:** The number of bits required to store a pixel of an image.
- ❑ **Aspect Ratio:** The ratio of an image width to its height, measured in unit length or number of pixels, is referred to as its aspect ratio. A 1024×768 image has an aspect ratio of $4/3$.

Spatial Resolution (cont...)



1024



512



256



128



64

32



Typical effects of
reducing spatial
resolution.

Images shown at
(a) 1250 ppi,
(b) 300 ppi,
(c) 150 ppi and
(d) 72 ppi

Intensity Level Resolution

Intensity level resolution refers to the number of intensity levels used to represent the image

- The more intensity levels used, the finer the level of detail in an image
- Intensity level resolution is usually given in terms of the number of bits used to store each intensity level

Number of Bits	Number of Intensity Levels	Examples
1	2	0, 1
2	4	00, 01, 10, 11
4	16	0000, 0101, 1111
8	256	00110011, 01010101
16	65,536	1010101010101010

Intensity Level Resolution (cont...)

256 grey levels (8 bits per pixel)



128 grey levels (7 bpp)



64 grey levels (6 bpp)



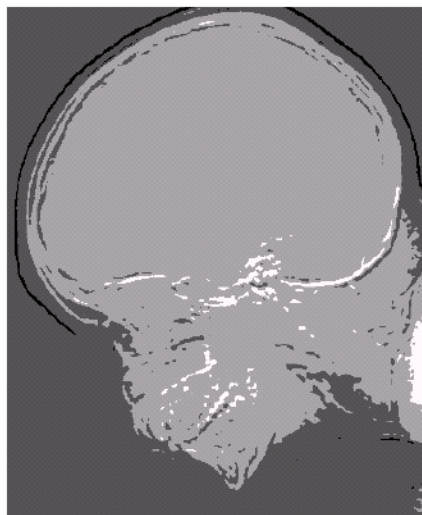
32 grey levels (5 bpp)



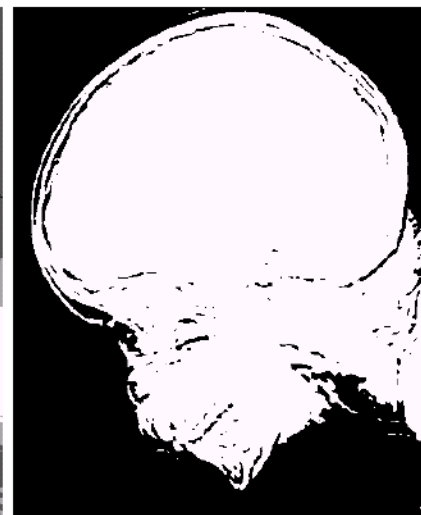
16 grey levels (4 bpp)



8 grey levels (3 bpp)



4 grey levels (2 bpp)



2 grey levels (1 bpp)

Some Basic Relationships between Pixels

- A pixel p at location (x, y) has **8-neighbors** comprise **four neighbors (horizontal and vertical)** at the locations:

$$(x + 1, y), (x - 1, y), (x, y + 1) \text{ and } (x, y - 1)$$

- and **four diagonal neighbors** at the locations:

$$(x + 1, y + 1), (x - 1, y - 1), (x - 1, y + 1) \text{ and } (x + 1, y - 1).$$

- $N_4(p) \rightarrow 4 - \text{neighbors of pixel } p$
- $N_D(p) \rightarrow 4 - \text{diagonal neighbors of pixel } p$
- $N_8(p) \rightarrow 8 - \text{neighbors of pixel } p \left(N_4(p) \cup N_D(p) \right)$

$(x - 1, y - 1)$	$(x - 1, y)$	$(x - 1, y + 1)$
$(x, y - 1)$	(x, y)	$(x, y + 1)$
$(x + 1, y - 1)$	$(x + 1, y)$	$(x + 1, y + 1)$

**8-neighborhood of pixel
 (x, y)**

- Let V be the set of gray-level values used to define adjacency.
- In a binary image, $V = \{1\}$ if we are referring to adjacency of pixels with value 1.
- In a gray scale image, the idea is the same, but set V typically contains more elements. For example, in the adjacency of pixels with a range of possible gray-level values 0 to 255, set V could be any subset of these 256 values.

The three types of adjacency:

- **4-Adjacency:** Two pixels p and q with values from V are adjacent if q is in the set $N_4(p)$. (Here V is the set of gray values $V = \{V_1, V_2, \dots, V_n\}$, where n is less than or equal to the total number of gray levels).

p and q are 4 – adjacent if $q \in N_4(p)$, and $p, q \in V$

- **8-Adjacency:** Two pixels p and q with values from V are adjacent if q is in the set $N_8(p)$.

p and q are 8 – adjacent if $q \in N_8(p)$, and $p, q \in V$

where $N_8(p) = N_4(p) \cup N_D(p)$.

- **m-Adjacency (Mixed Adjacency):** Two pixels p and q with values from V are m-adjacent if

➤ *q is in $N_4(p)$, or*

➤ *q is in $N_D(p)$ and the set $N_4(p) \cap N_4(q) = \emptyset$,*

(i.e. the set $N_4(p) \cap N_4(q)$ has no pixel whose values are from V)

- Mixed adjacency** is a modification of 8-adjacency. This is used to eliminate the ambiguities that often arise when 8-adjacency is used. Two image subsets S_1 and S_2 are adjacent if some pixel in S_1 is adjacent to some pixel in S_2 .

0 1 1
0 1 0
0 0 1

(a)

Pixels
Arrangement

0 1---1
0 1---0
0 0 1

(b)

Pixels that are 8-Adjacent
to the Central Pixel.

0 1---1
0 1---0
0 0 1

(c)

m-Adjacency

- The top 3 pixels in (b) show multiple (ambiguous) 8-adjacency, as indicated by the dashed lines. The ambiguity is removed by using m-adjacency in (c).

- A **(digital) path** or curve from pixel p with coordinate (x, y) to pixel q with coordinate (s, t) is a sequence of distinct pixels with coordinates

$$(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n).$$

where $(x_0, y_0) \equiv (x, y)$ *and* $(x_n, y_n) \equiv (s, t)$, and pixels (x_i, y_i) are adjacent for $1 \leq i \leq n - 1$. In this case n is the length of the path. If $(x_0, y_0) \equiv (x_n, y_n)$, then the path is a closed path. We can define 4-, 8-, or m -paths depending upon the type of adjacency specified.

- Let S represent a subset of pixels in an image. Two pixels p and q are said to be **connected** in S if there exists a path between them consisting entirely of pixels in S .
- For any pixel p in S , the set of pixels that are connected to it in S is called a **connected component**. If it has only one connected component, then set S is called a **connected set**.

- For pixels p , q and z , with coordinates (x, y) , (s, t) and (v, w) , D is a distance function or metric if
 - ❖ $D(p, q) \geq 0$ ($D(p, q) = 0$, if $p = q$)
 - ❖ $D(p, q) = D(q, p)$
 - ❖ $D(p, z) \leq D(p, q) + D(q, z)$

1. The **Euclidean distance** between p and q is defined as:

$$D_e(p, q) = \sqrt{(x - s)^2 + (y - t)^2}$$

For this distance measure, the pixels having a distance less than or equal to some value r from (x, y) are the points contained in a **disk** of radius r , centered at (x, y) .

2. The **D_4 distance** (also called **city-block distance**) between p and q is defined as:

$$D_4(p, q) = |x - s| + |y - t|$$

In this case, the pixels with D_4 distance from (x, y) less than or equal to some value r form a **diamond** centered at (x, y) . For example, the pixels with D_4 distance ≤ 2 from (x, y) (the center point) form the following

$$\begin{array}{ccccc} & & 2 & & \\ & 2 & 1 & 2 & \\ 2 & 1 & 0 & 1 & 2 \\ & 2 & 1 & 2 & \\ & & 2 & & \end{array}$$

The pixels with $D_4 = 1$ are the 4-neighbors of (x, y) .

3. The **D_8 distance** (also called **chessboard distance**) between p and q is defined as:

$$D_8(p, q) = \max(|x - s|, |y - t|)$$

In this case, the pixels with D_8 distance from (x, y) less than or equal to some value r form a **square** centered at (x, y) . For example, the pixels with D_8 distance ≤ 2 from (x, y) (the center point) form the following

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2

- The pixels with $D_8 = 1$ are the 8-neighbors of (x, y) .

- Let $V = \{1\}$ and $p = p_2 = p_4 = 1$
- Assume pixel p as source and pixel p_4 as destination.

Case 1: If $p_1 = p_3 = 0$,

$m - path$	$p \rightarrow p_2 \rightarrow p_4$
D_m	2

	p_3	p_4
p_1	p_2	
p		

Case 2: If $p_1 = p_3 = 1$,

$m - path$	$p \rightarrow p_1 \rightarrow p_2 \rightarrow p_3 \rightarrow p_4$
D_m	4

Image Processing Operations

Divided into three categories

❖ **Point Processing Operations**

$$Output(x, y) \propto Input(x_0, y_0)$$

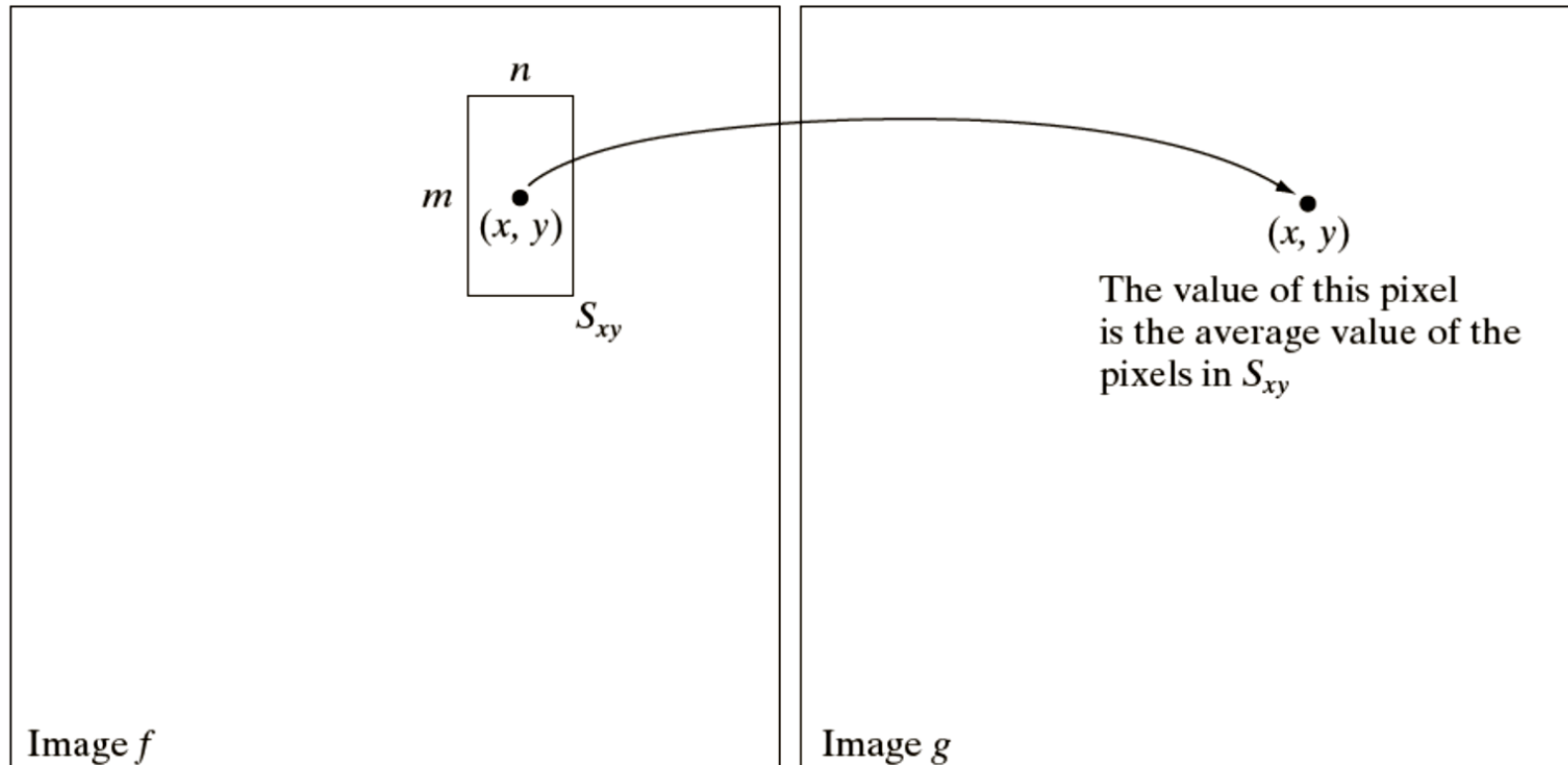
❖ **Local Operations**

$$Output(x, y) \propto Neighborhood\ of\ Input(x_0, y_0)$$

❖ **Global Operations**

$$Output(x, y) \propto All\ pixels\ of\ Input\ Image$$

Image Processing Operations



Arithmetic Operations

- ❑ Arithmetic operations are point operations applied between two images of equal size.
- ❑ They include following operations:
 - ❖ Image Addition
 - ❖ Image Subtraction
 - ❖ Image Multiplication
 - ❖ Image Division

Arithmetic Operations: Image Addition

- ❑ The output image is formed by the addition of two input images or addition of an image with some constant K

$$g(x, y) = f_1(x, y) + f_2(x, y)$$

OR

$$g(x, y) = f_1(x, y) + K$$

Applications

- ❑ **To increase the brightness of the image.**

The brightness of an image is the average pixel intensity of an image. If a positive or negative constant is added to all the pixels of an image, the average intensity of the image increases or decreases.

Arithmetic Operations: Image Addition



Original Image

+ 50 =



K

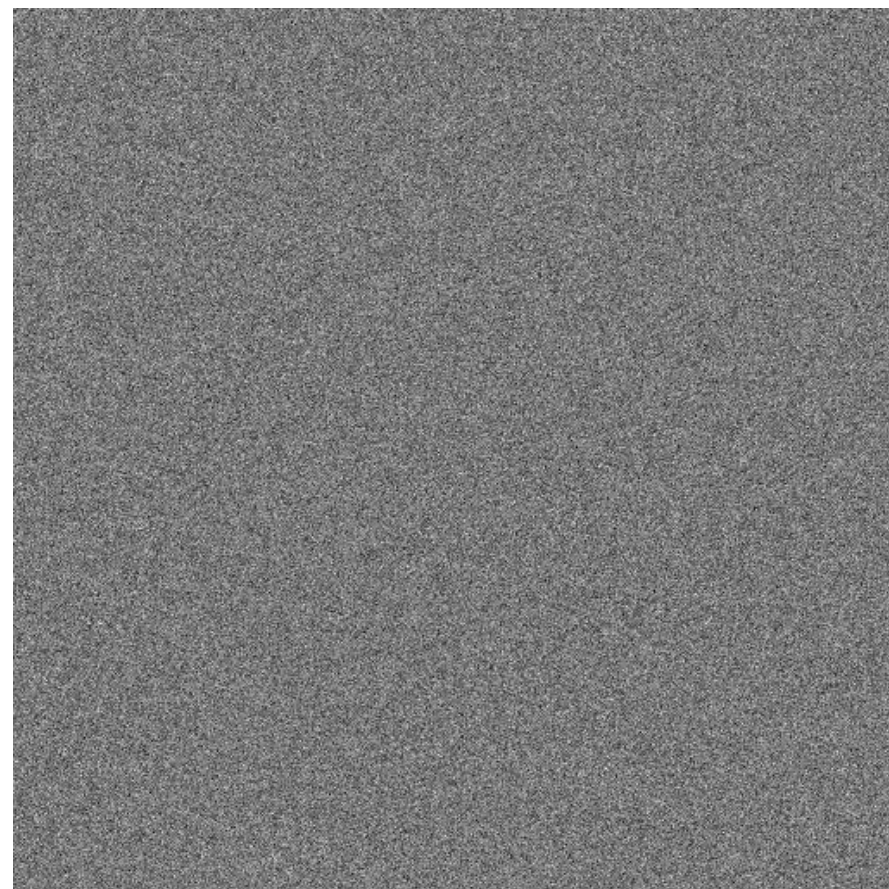


Output Image

Arithmetic Operations: Image Addition



+



Original Image $f_1(x, y)$

Noise $f_2(x, y)$

Arithmetic Operations: Image Addition



Output Image
 $g(x, y)$

Arithmetic Operations: Image Subtraction

- ❑ The output image is formed by the subtraction of two input images or subtraction of an image with some constant K

$$g(x, y) = f_1(x, y) - f_2(x, y)$$

OR

$$g(x, y) = f_1(x, y) - K$$

Applications

- ❑ **To decrease the brightness of the image.**
- ❑ **Motion or Change Detection in video sequences.**

Arithmetic Operations: Image Subtraction



Original Image

− 50 =



K



Output Image

Arithmetic Operations: Image Subtraction



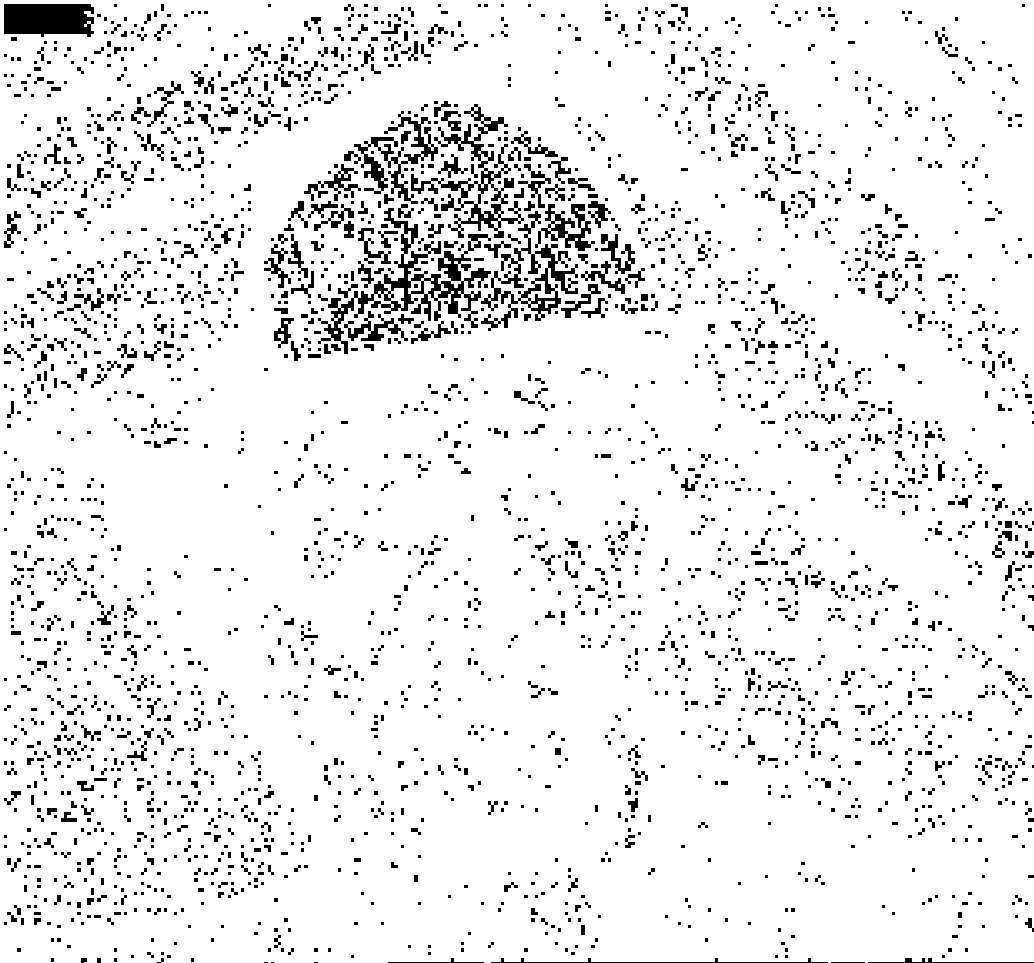
Frame 1 $f_1(x, y)$

—



Frame 2 $f_2(x, y)$

Arithmetic Operations: Image Subtraction



Output Image
 $g(x, y)$

Arithmetic Operations: Image Multiplication

- ❑ The output image is formed by the point (or pair) wise multiplication of two input images or multiplication of an image with some constant K

$$g(x, y) = f_1(x, y) \times f_2(x, y)$$

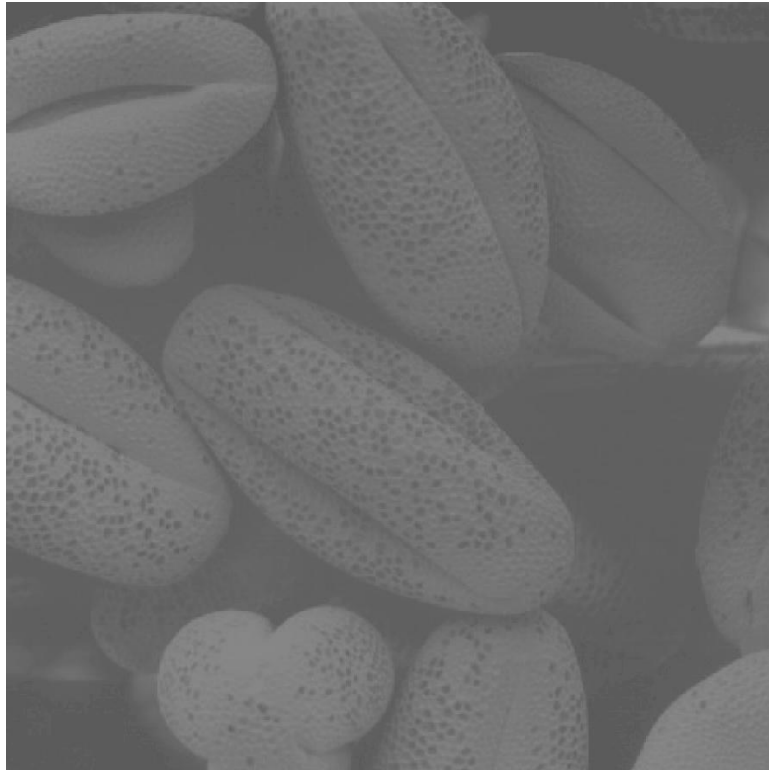
OR

$$g(x, y) = f_1(x, y) \times K$$

Applications

- ❑ **To increase the contrast of the image.**
- ❑ **It is useful for extracting the area of interest in an image.**

Arithmetic Operations: Image Multiplication

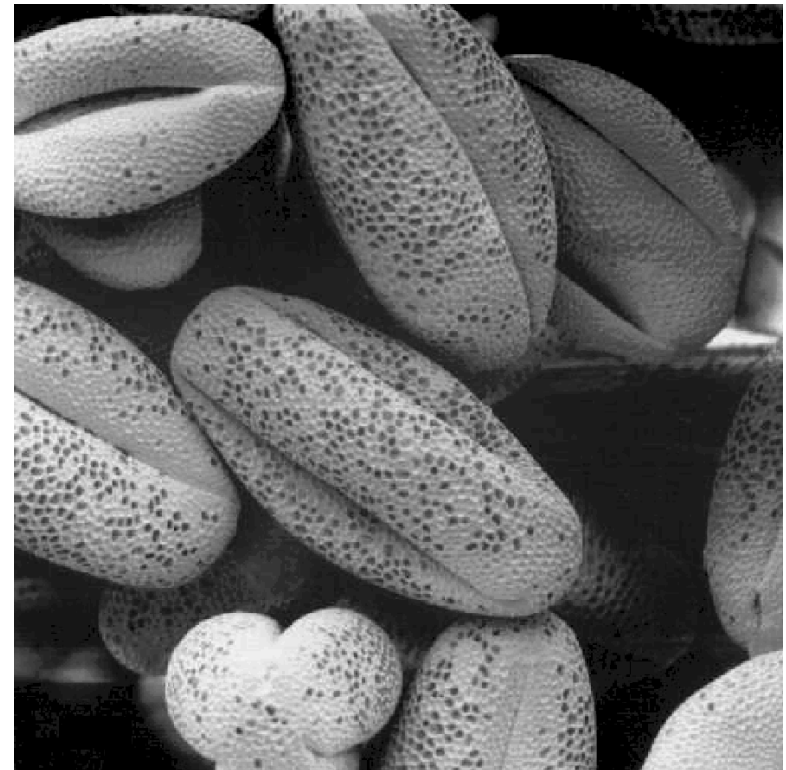


Original Image

$\times 2 =$

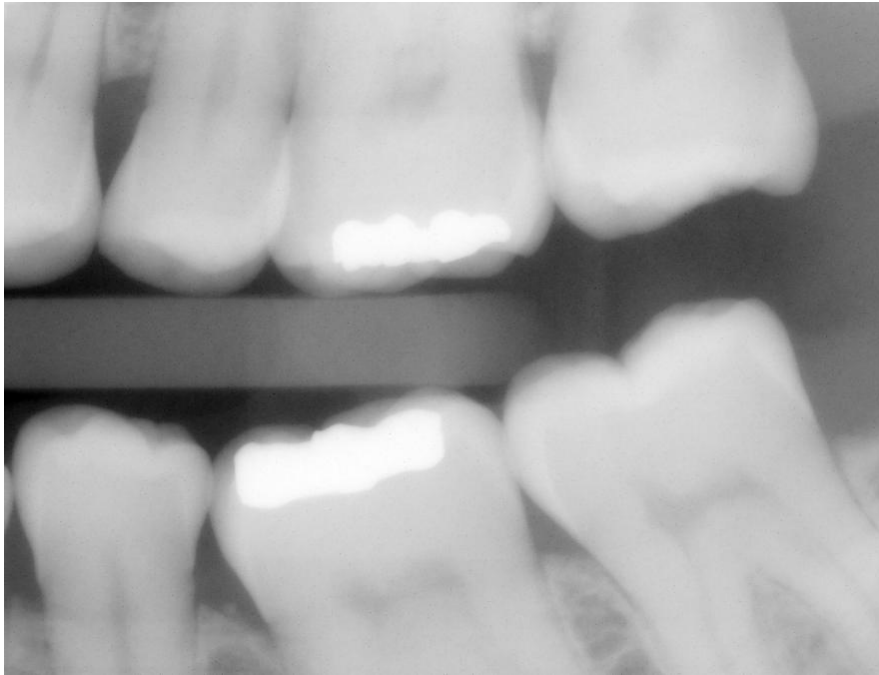


K



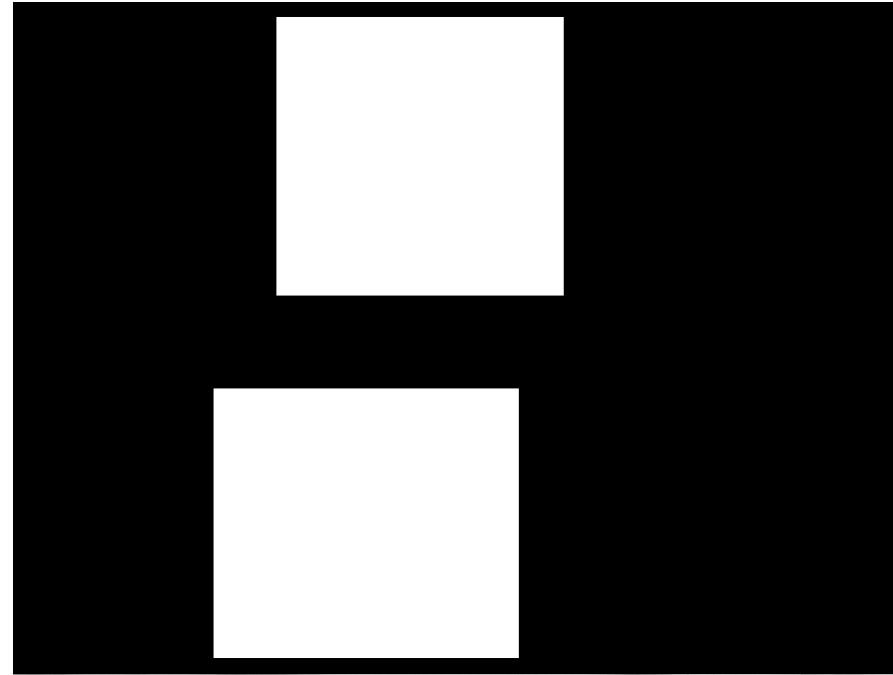
Output Image

Arithmetic Operations: Image Multiplication



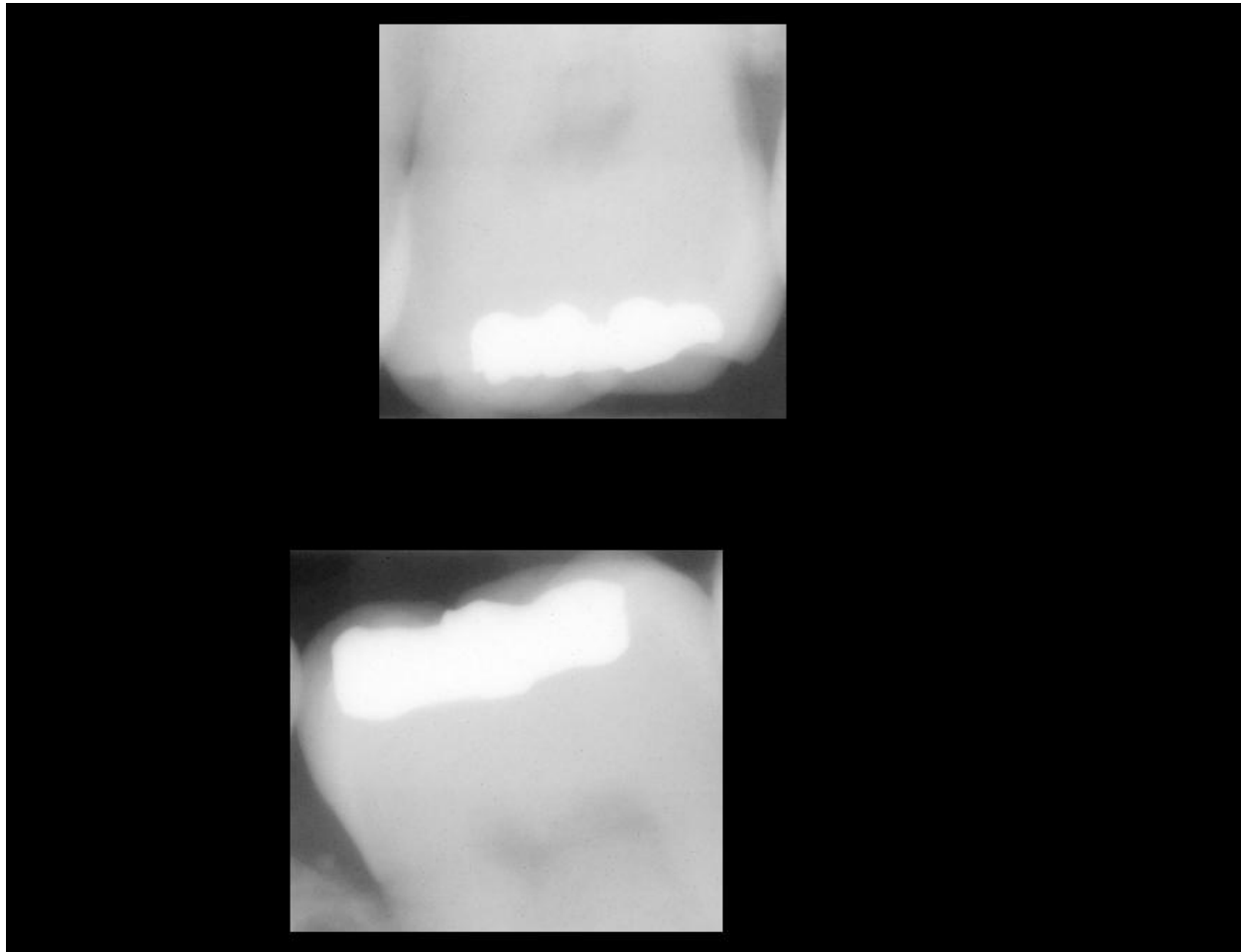
Original Image $f_1(x, y)$

×



Binary Mask $f_2(x, y)$

Arithmetic Operations: Image Multiplication



Output Image
 $g(x, y)$

Arithmetic Operations: Image Division

- ❑ The output image is formed by the point (or pair) wise division of two input images or division of an image with some constant K

$$g(x, y) = f_1(x, y) / f_2(x, y)$$

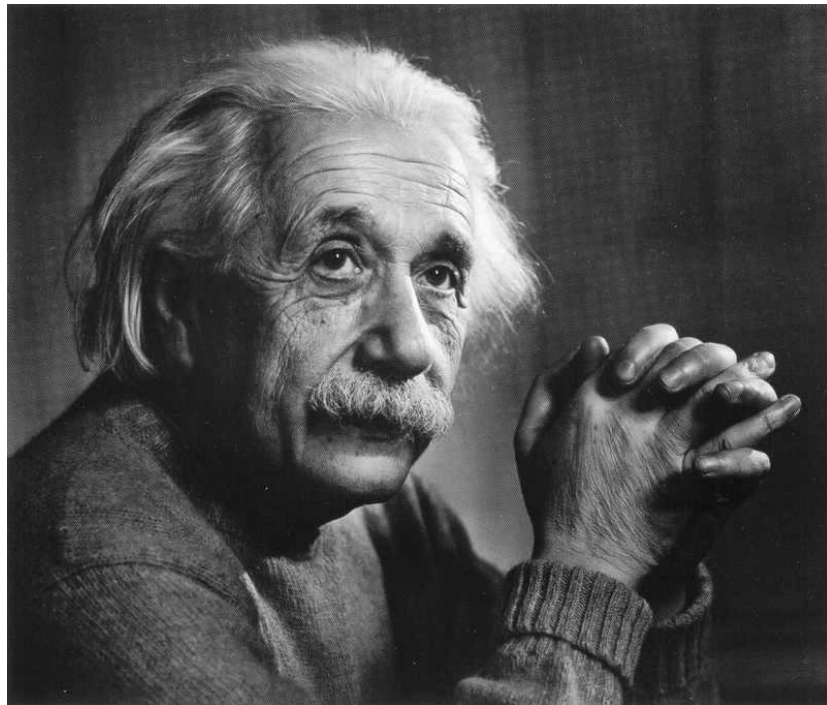
OR

$$g(x, y) = f_1(x, y) / K$$

Applications

- ❑ **To decrease the contrast of an image**

Arithmetic Operations: Image Division

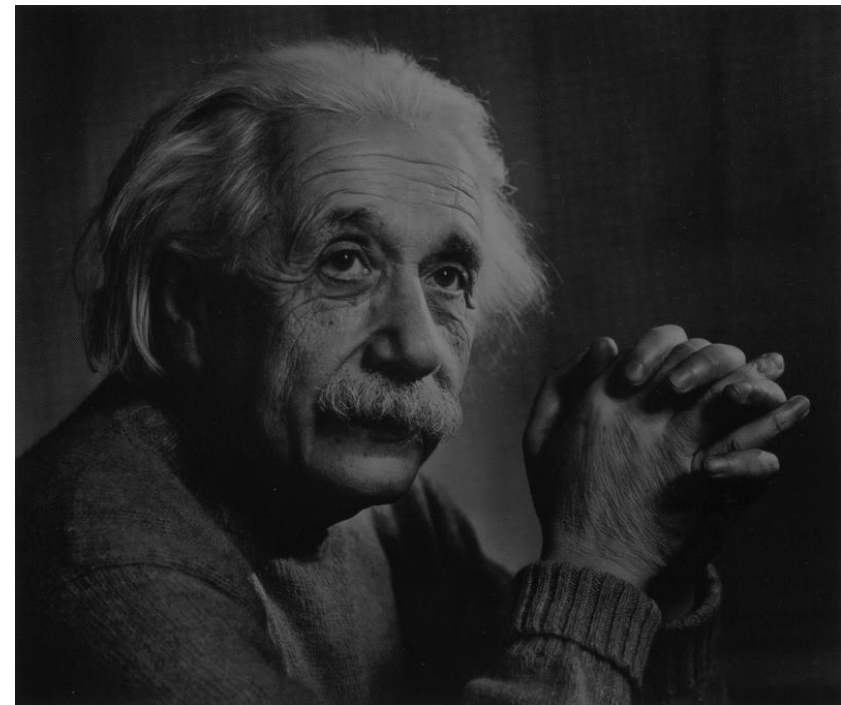


Original Image

$/ 2 =$



K



Output Image

Arithmetic Operations: Image or Alpha Blending

- ❑ This operation blends two images of same size to yield a resultant image. This can be mathematically written as:

$$g(x, y) = \alpha \times f_1(x, y) + (1 - \alpha) \times f_2(x, y)$$

- ❑ α is called blending ratio, which determines the influence of each image on the resultant image.

Arithmetic Operations: Image or Alpha Blending



Image 1 $f_1(x, y)$

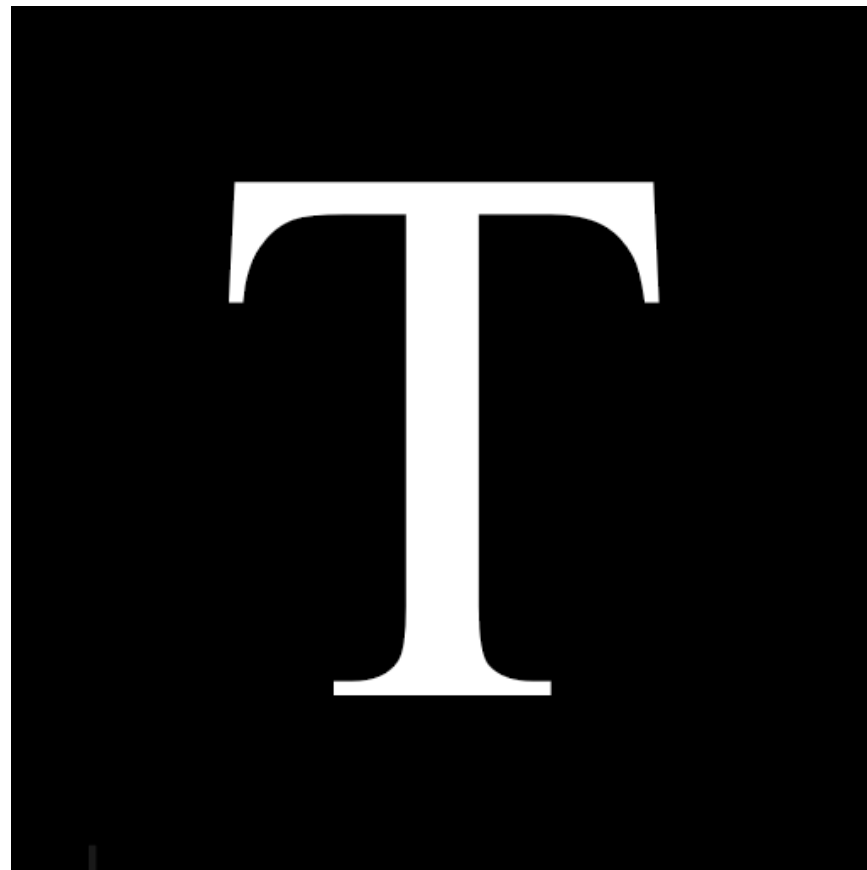


Image 2 $f_2(x, y)$

Arithmetic Operations: Image or Alpha Blending



**Blended Output
Image $g(x, y)$**

Arithmetic Operations: Image or Alpha Blending



Image 1 $f_1(x, y)$



Image 2 $f_2(x, y)$

Arithmetic Operations: Image or Alpha Blending



**Blended Output
Image $g(x, y)$**

- ❑ Arithmetic operations are point operations applied between two images of equal size.
- ❑ They include following operations:
 - ❖ OR/UNION ($\cup$)
 - ❖ AND/INTERSECTION ($\cap$)
 - ❖ NOT/COMPLEMENT
- ❑ These operators are widely used in morphological image processing such as image thinning and thickening, boundary extraction, extraction of connected components etc.

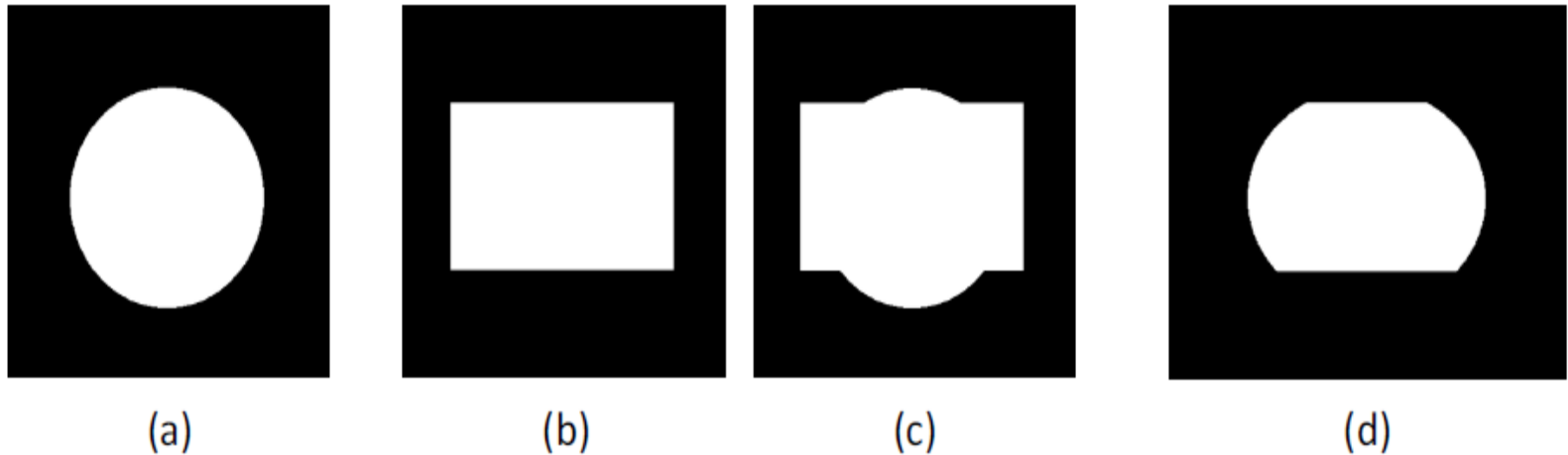


Fig. 3.18 Results of the AND and OR logical operators (a) Image 1 (b) Image 2
(c) Result of image 1 OR image 2 (d) Result of image 1 AND image 2

- Image Interpolation is a basic tool used extensively in **Zooming, Shrinking, Rotating, and Geometric Corrections**.
- **Interpolation is the process of using known data to estimate values at unknown locations.**
- Many interpolation schemes are available, however, we will study two basic interpolations schemes namely:
 - ❖ **Nearest Neighbor Interpolation**
 - ❖ **Bilinear Interpolation**
- For a given image, if scaling factor (s)
 - $s > 1 \rightarrow$ **Image Zooming**
 - $s < 1 \rightarrow$ **Image Shrinking**
 - $s = 1 \rightarrow$ **No change in size**

Image Interpolation

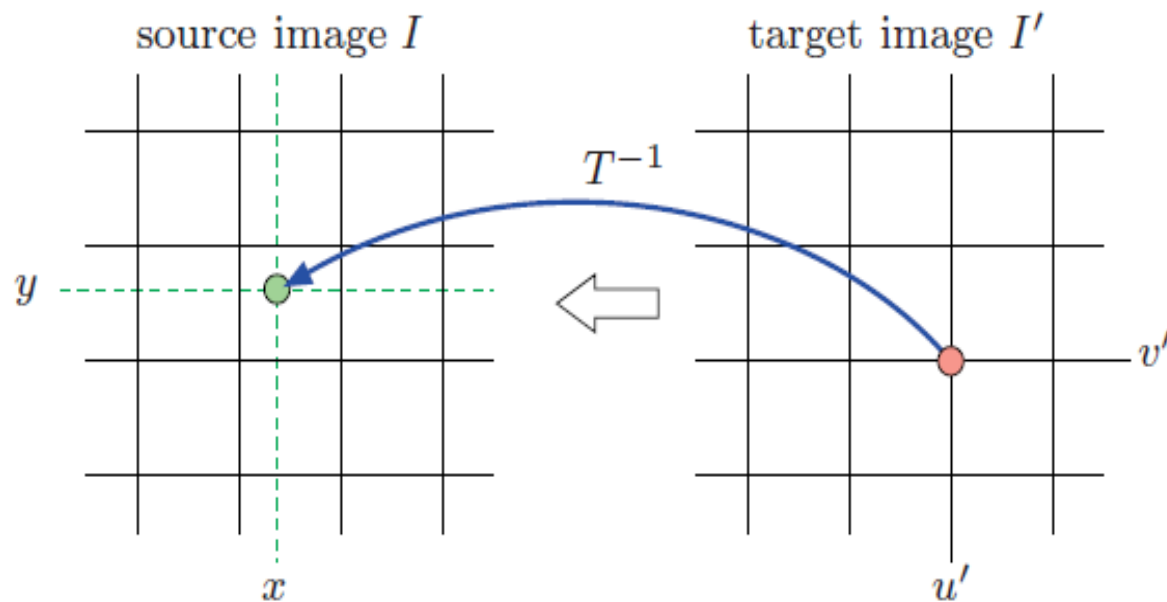


Figure 10.10 Target-to-source mapping. For each discrete pixel position (u', v') in the target image I' , the corresponding continuous source position (x, y) is found by applying the inverse mapping function $T^{-1}(u', v')$. The new pixel value $I'(u', v')$ is determined by interpolating the pixel values in the source image within some neighborhood of (x, y) .

- **Nearest Neighbor Interpolation**

It assigns to each new location the intensity of its nearest neighbor in the original image.

For example, let us assume that we want to interpolate the value of pixel at location (x, y) in the target image I' .

Step 1:

$$x_0 = T^{-1}(x) = x/s, \quad y_0 = T^{-1}(y) = y/s$$

Step 2:

$$u_0 = \lfloor x_0 \rfloor, \quad v_0 = \lfloor y_0 \rfloor$$

Step 3:

$$I'(x, y) = I(x_0, y_0) = I(u_0, v_0)$$

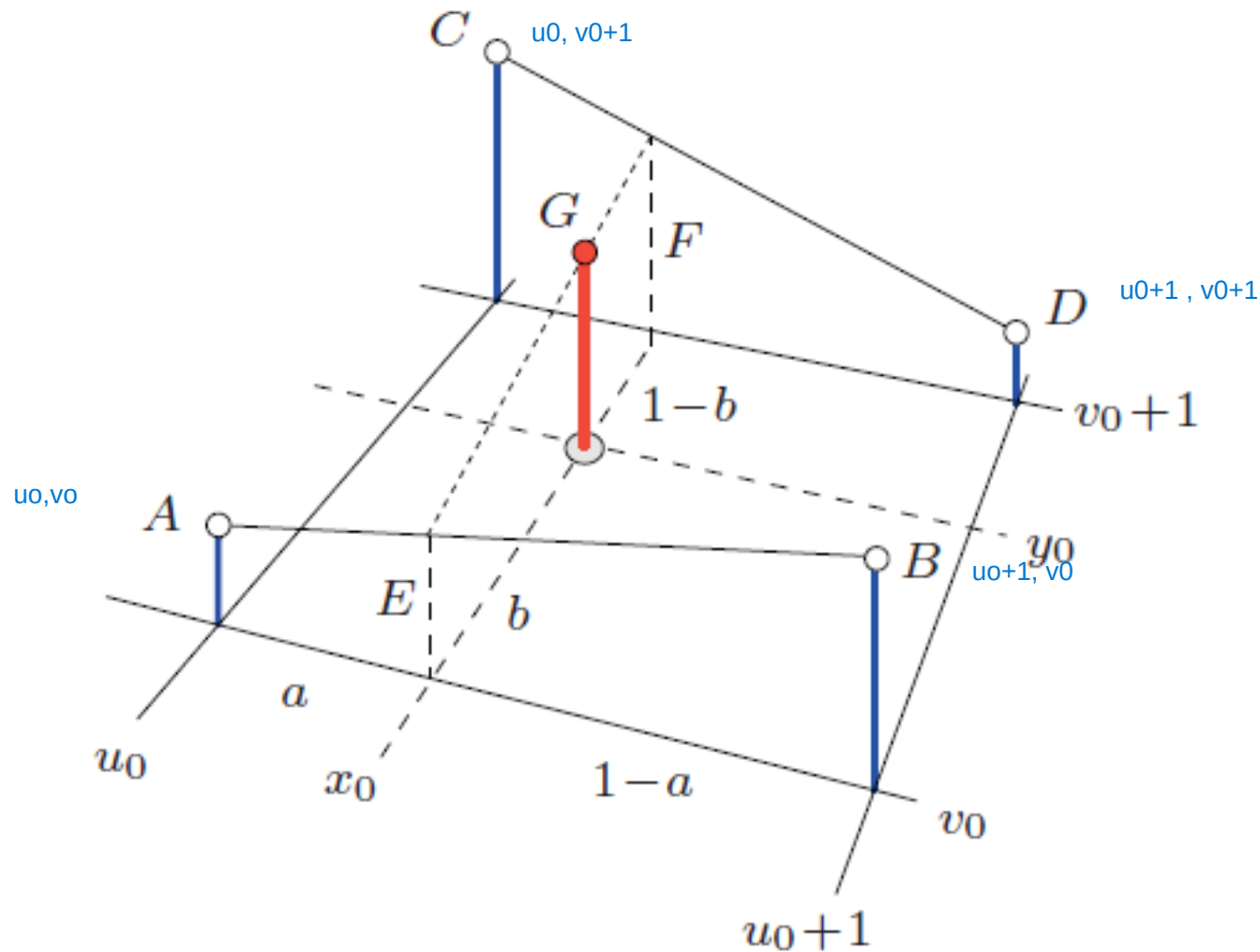
Bilinear Image Interpolation

- ❑ It uses four nearest neighbors to estimate the intensity at a given location.
- ❑ It uses ***linear interpolation*** to estimate the values of pixels along horizontal and vertical direction.

x	y
0.5	2
1	4
1.5	6
2	8
2.5	10
3	12
2.2	?

$$(y - y_1) = \frac{(y_2 - y_1)}{(x_2 - x_1)} (x - x_1)$$

Bilinear Image Interpolation



Bilinear Image Interpolation

- **Bilinear Interpolation**

It uses four nearest neighbors to estimate the intensity at a given location.

Step 1:	$x_0 = T^{-1}(x) = x/s, \quad y_0 = T^{-1}(y) = y/s$
Step 2:	$u_0 = \lfloor x_0 \rfloor, \quad v_0 = \lfloor y_0 \rfloor$
Step 3:	$\begin{aligned} A &= I(u_0, v_0), & B &= I(u_0 + 1, v_0) \\ C &= I(u_0, v_0 + 1), & D &= I(u_0 + 1, v_0 + 1) \end{aligned}$
Step 4:	$\begin{aligned} I'(x, y) &= (a - 1)(b - 1)A + a(1 - b)B + (1 - a)bC + abD \\ a &= x_0 - u_0, & b &= y_0 - v_0 \end{aligned}$

Bilinear Image Interpolation

$$E = A + (x_0 - u_0) \cdot (B - A) = A + a \cdot (B - A),$$

$$F = C + (x_0 - u_0) \cdot (D - C) = C + a \cdot (D - C),$$

$$I'(x, y) = I(x_0, y_0) = G$$

$$G = E + (y_0 - v_0) \cdot (F - E) = E + b \cdot (F - E)$$

$$I'(u, v) = (a - 1)(b - 1)A + a(1 - b)B + (1 - a)bC + abD$$

$$a = x_0 - u_0, \quad b = y_0 - v_0$$

Image Interpolation

- The nearest neighbor interpolation is fast, however, the quality of interpolation is not good. It provides a checkerboard (blocky) effect that is particularly objectionable at high degree of magnification.
- Bilinear interpolation gives better results than the nearest neighbor interpolation. It does a better job of preserving fine detail.
- Higher the number of neighbors involved in the interpolation scheme, the better will be the estimated value.

